Cosserat Rod Model for a Concentric Tube Robot

A continuum-mechanics treatment of CTR shape, specialised at the end to the case of one rigid straight outer tube and one pre-curved flexible inner tube.

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1. Why a continuum model?

A CTR is not a finite-DOF kinematic chain — it is a slender elastic structure whose shape is a curve in $\mathbb{R}^3$ governed by elasticity. The 15-segment URDF used earlier is a discrete approximation that converges to the true continuum solution in the limit of infinite segments. The standard continuum model for thin rods that bend AND twist is the **Cosserat rod**.

Notation. Throughout, primes denote d/ds where s is arc length. Vectors are columns; $[\mathbf{a}]_\times$ is the skew-symmetric matrix such that $[\mathbf{a}]_\times \mathbf{b} = \mathbf{a} \times \mathbf{b}$. Frames are right-handed.

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2. The Cosserat rod

2.1 Configuration

Parameterise the rod by arc length $s \in [0, L]$. At every s there are two state quantities:

- a **centerline position** $\mathbf{p}(s) \in \mathbb{R}^3$,
- a **cross-section orientation** $\mathbf{R}(s) \in SO(3)$,

so the configuration is a curve $g(s) = (\mathbf{R}(s), \mathbf{p}(s)) \in SE(3)$. The columns of $\mathbf{R}(s)$ are the body-frame axes $(\hat{\mathbf{e}}_x, \hat{\mathbf{e}}_y, \hat{\mathbf{e}}_z)$ at s; by convention the body $\hat{z}$-axis points along the rod tangent.

The defining feature of the Cosserat model (vs. simple Frenet–Serret) is that $\mathbf{R}(s)$ is independent of $\mathbf{p}'(s)$ — cross-sections may rotate without translating. This lets the model carry shear and twist as independent kinematic states.

2.2 Body-frame strains

Take derivatives:

$$\mathbf{p}'(s) = \mathbf{R}(s)\,\mathbf{v}(s), \qquad \mathbf{R}'(s) = \mathbf{R}(s)\,[\mathbf{u}(s)]_\times$$

where the **body-frame strain vectors** are

$$\mathbf{v}(s) = \begin{pmatrix} v_x \\ v_y \\ v_z \end{pmatrix} \quad \text{(linear)}, \qquad \mathbf{u}(s) = \begin{pmatrix} u_x \\ u_y \\ u_z \end{pmatrix} \quad \text{(angular)}.$$

Components have direct physical meaning:

Component	Meaning
v_x, v_y	shear
v_z	axial stretch ($v_z = 1$ unstretched)
u_x, u_y	bending curvatures about local $\hat{x}, \hat{y}$
u_z	torsion (twist rate)

These two ODEs are the **kinematic relations**: given $(\mathbf{v}, \mathbf{u})$ along the rod, integrating them from a base pose reconstructs the entire shape.

2.3 Constitutive law

The rod has a **natural** (stress-free) reference state with strains $\mathbf{v}^*(s), \mathbf{u}^*(s)$. Linear elasticity then gives the body-frame internal force $\mathbf{N}$ and moment $\mathbf{M}$:

$$\mathbf{N}(s) = \mathbf{K}_{se}\left(\mathbf{v}(s) - \mathbf{v}^*(s)\right), \qquad \mathbf{M}(s) = \mathbf{K}_{bt}\left(\mathbf{u}(s) - \mathbf{u}^*(s)\right)$$

For an isotropic circular cross-section,

$$\mathbf{K}_{se} = \operatorname{diag}(GA,\ GA,\ EA), \qquad \mathbf{K}_{bt} = \operatorname{diag}(EI,\ EI,\ GJ)$$

with E Young's modulus, $G = E/[2(1+\nu)]$ shear modulus, A section area, I second moment, $J = 2I$ polar second moment. For a hollow circular section with inner/outer radii r_i, r_o:

$$A = \pi(r_o^2 - r_i^2), \qquad I = \tfrac{\pi}{4}(r_o^4 - r_i^4)$$

The **precurvature** $\mathbf{u}^*(s)$ encodes the rod's intrinsic shape memory. For your inner tube — a planar arc of radius R in the body xz-plane — the precurvature is constant along the rod and points along body $\hat{y}$:

$$\mathbf{u}^*_{\text{inner}} = \begin{pmatrix} 0 \\ \kappa^* \\ 0 \end{pmatrix}, \qquad \kappa^* = \frac{1}{R} = 20\,\text{m}^{-1}.$$

For an unstretched inextensible rod, $\mathbf{v}^* = (0, 0, 1)^\top$.

2.4 Equilibrium

Per-unit-length balance of forces and moments in the world frame, with distributed external load $\mathbf{f}(s)$ and moment $\mathbf{l}(s)$:

$$\mathbf{n}'(s) + \mathbf{f}(s) = \mathbf{0}$$

$$\mathbf{m}'(s) + \mathbf{p}'(s) \times \mathbf{n}(s) + \mathbf{l}(s) = \mathbf{0}$$

with world-frame internals $\mathbf{n} = \mathbf{R}\mathbf{N}, \mathbf{m} = \mathbf{R}\mathbf{M}$.

Together the kinematic, constitutive, and equilibrium relations give a system of ODEs in $(\mathbf{p}, \mathbf{R}, \mathbf{n}, \mathbf{m})$ — a boundary value problem once you supply pose at $s = 0$ and tip wrench at $s = L$.

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3. The Kirchhoff simplification

CTRs are very slender (millimetre diameters, centimetre lengths). Axial stretch and shear are negligible compared to bending and torsion. The **Kirchhoff rod** is the Cosserat rod with the rigid constraint

$$\mathbf{v}(s) \equiv (0, 0, 1)^\top.$$

Consequences:

- The tangent equals the body $\hat{z}$-axis: $\mathbf{p}'(s) = \mathbf{R}(s)\hat{z}$.
- Only $\mathbf{u}(s)$ varies; only $\mathbf{K}_{bt}$ is needed.
- $\mathbf{N}(s)$ becomes a Lagrange multiplier enforcing $\mathbf{v} = \hat{z}$ (determined by the equilibrium equations rather than by a constitutive law).

The rod's elastic energy reduces to the bending–torsion form

$$U[\mathbf{u}] = \tfrac{1}{2}\int_0^L \big(\mathbf{u}(s) - \mathbf{u}^*(s)\big)^\top \mathbf{K}_{bt}\big(\mathbf{u}(s) - \mathbf{u}^*(s)\big)\, ds.$$

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4. Multiple concentric tubes

Now the CTR-specific structure. With n tubes nested concentrically at common arc length s, the standard assumption is that they share

- a common centerline $\mathbf{p}(s)$,
- a common bending frame $\mathbf{R}(s)$,

but each tube can rotate independently about the local $\hat{z}$ by an angle $\theta_i(s)$. So tube i's body frame is

$$\mathbf{R}_i(s) = \mathbf{R}(s)\, R_z(\theta_i(s)).$$

4.1 Tube-local curvature

Differentiating $\mathbf{R}_i(s) = \mathbf{R}(s) R_z(\theta_i(s))$ and reading off the body-frame angular velocity gives

$$\mathbf{u}^{(i)}(s) = R_z(-\theta_i(s))\, \mathbf{u}(s) + \theta_i'(s)\, \hat{z}.$$

Component-wise,

$$\begin{aligned}
u^{(i)}_x &= u_x \cos\theta_i + u_y \sin\theta_i \\
u^{(i)}_y &= -u_x \sin\theta_i + u_y \cos\theta_i \\
u^{(i)}_z &= u_z + \theta_i'.
\end{aligned}$$

Each tube has its own precurvature $\mathbf{u}_i^*$ in its own body frame.

4.2 The bending compromise

The total elastic energy is

$$U = \tfrac{1}{2}\sum_i \int \big(\mathbf{u}^{(i)} - \mathbf{u}_i^*\big)^\top \mathbf{K}_{bt,i}\big(\mathbf{u}^{(i)} - \mathbf{u}_i^*\big)\, ds.$$

Stationarity in the common bending components u_x, u_y at each s yields the **stiffness-weighted average** of the precurvatures:

$$u_x(s) = \frac{\sum_i E_i I_i \left(u_{i,x}^* \cos\theta_i + u_{i,y}^* \sin\theta_i\right)}{\sum_i E_i I_i}$$

$$u_y(s) = \frac{\sum_i E_i I_i \left(-u_{i,x}^* \sin\theta_i + u_{i,y}^* \cos\theta_i\right)}{\sum_i E_i I_i}$$

Physically: each tube tries to enforce its own precurvature; they compromise on a curve weighted by their bending stiffness. As the tubes are rotated relative to one another (the user changes θ_i), the resulting curvature changes — that is the **steering principle** of a CTR.

4.3 Torsional ODE

For each tube, balancing the torsional moment along the backbone yields

$$\theta_i''(s) = \frac{E_i I_i}{G_i J_i} \left(u_x\, u_{i,y}^* - u_y\, u_{i,x}^*\right)$$

(using a common sign convention; primary references differ by signs depending on rotation conventions). With prescribed $\theta_i(0)$ and free-tip $\theta_i'(L_i) = 0$, this is a BVP per tube, coupled to the bending equations through the θ_i in u_x, u_y.

These coupled equations — bending compromise plus per-tube torsional ODE — constitute the **Webster–Dupont–Rucker model** of CTR statics (see Rucker, Jones, Webster, IROS 2010 / IJRR 2011).

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5. Your case: rigid outer + flexible inner, no external load

Now collapse the framework to two tubes:

- Tube 1 (outer): rigid, straight, length $L_o = 30\,\text{mm}$. Effectively $E_1 I_1 \to \infty$, $\kappa_1^* = 0$.
- Tube 2 (inner): flexible, $\kappa_2^* = 1/R = 20\,\text{m}^{-1}$, length L_i. User controls insertion d (protrusion past the outer tube tip) and base rotation ψ.

5.1 Inside the outer tube

Taking $E_1 I_1 \to \infty$ in the bending compromise drives the weighted average to the rigid tube's precurvature:

$$u_x \to u^*_{1,x} = 0, \qquad u_y \to u^*_{1,y} = 0.$$

The backbone is forced straight inside the outer tube. The inner tube is then geometrically constrained to the same straight centerline; the difference between its actual curvature ($\mathbf{u}^{(2)} = \theta_2' \hat{z}$) and its precurvature ($\mathbf{u}_2^* = \kappa^* \hat{y}$) is

$$\mathbf{u}^{(2)} - \mathbf{u}_2^* = (0,\ -\kappa^*,\ \theta_2').$$

The inner tube therefore carries a **constant bending moment**

$$M_{\text{bend}} = E_2 I_2\, \kappa^*$$

stored as elastic strain energy along the entire constrained portion. This moment is reacted by contact forces between the inner and outer tube walls — it does NOT bend the centerline, because the rigid outer tube absorbs it. (You can think of the rigid outer tube as enforcing a Lagrange-multiplier constraint $\mathbf{u}_\perp = \mathbf{0}$ along its length, with the contact reaction being the corresponding multiplier.)

In the torsion direction, with no friction, the moment-balance equation gives $\theta_2''(s) = 0$ inside the outer tube — uniform twist rate. With the boundary $\theta_2(0) = \psi$ at the base and continuity at the outer tube's exit, θ_2 rotates linearly along the constrained portion or stays constant if no torque is applied.

5.2 Outside the outer tube — the protruding portion

For arc length past the outer tube's tip, $s \in [L_o,\ L_o + d]$, the inner tube is unconstrained. With no external force or moment, the equilibrium ODEs $\mathbf{n}' = 0, \mathbf{m}' = 0$ admit the trivial solution $\mathbf{n} \equiv 0, \mathbf{m} \equiv 0$. Constitutively, this forces

$$\mathbf{u}(s) = \mathbf{u}_2^*(s) \quad \Longleftrightarrow \quad \text{the rod adopts its natural shape.}$$

The protruding portion is therefore a **circular arc of radius** R lying in a plane whose orientation is determined by $\theta_2(L_o)$ — the twist angle at the moment of exit. Without torsional friction or stored twist, $\theta_2(L_o) = \psi$, the user's input rotation.

5.3 Forward kinematics

Working in a world frame whose origin sits at the outer tube's base and whose $\hat{z}$ is along the outer tube axis:

Inside the outer tube ($0 \le s \le L_o$):

$$\mathbf{p}(s) = (0,\ 0,\ s)^\top, \qquad \mathbf{R}(s) = I.$$

At the exit ($s = L_o$): position $(0, 0, L_o)^\top$, frame rotated by ψ about the local $\hat{z}$:

$$\mathbf{R}(L_o) = R_z(\psi).$$

Protruding portion ($s = L_o + s', s' \in [0, d]$): planar circular arc of radius R in the local xz-plane, then rotated by ψ about the world $\hat{z}$. Letting $\varphi = s'/R \in [0,\ d/R]$:

$$\mathbf{p}(L_o + s') = \begin{pmatrix} 0 \\ 0 \\ L_o \end{pmatrix} + R_z(\psi) \begin{pmatrix} R(1 - \cos\varphi) \\ 0 \\ R\sin\varphi \end{pmatrix}$$

The tip ($s' = d, \varphi = d/R$):

$$\boxed{\mathbf{p}_{\text{tip}}(d, \psi) = \begin{pmatrix} 0 \\ 0 \\ L_o \end{pmatrix} + R_z(\psi) \begin{pmatrix} R(1 - \cos(d/R)) \\ 0 \\ R\sin(d/R) \end{pmatrix}}$$

This is exactly the geometry from the previous document — but now derived as the **trivial** ($\mathbf{n} = \mathbf{m} = 0$) **solution of the Cosserat BVP for the protruding portion**, not just postulated as a circular arc.

5.4 Where the discrete URDF model fits in

The 15-segment approximation in your URDF is the trapezoidal discretisation of the continuous arc. Each segment of length $\ell_s = d_{\text{full}}/M$ implements one Euler step of the orientation ODE $\mathbf{R}'(s) = \mathbf{R}(s)[\mathbf{u}^*]_\times$ with bend angle $\Delta\beta = \kappa^*\ell_s$. The piecewise-linear activation function in §6 of the geometry document is the discrete analogue of the continuous shape-emergence: as the rod is inserted by d, the continuum solution is a circular arc of length d, and the discrete model approximates this with $\lfloor d/\ell_s \rfloor$ fully-bent joints plus one partially-bent.

In the limit $M \to \infty$ the discrete model converges pointwise to the Cosserat solution.

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6. When the full machinery actually matters

For your specific setup the answer collapses because the outer tube is rigid and there are no external loads. The framework is ESSENTIAL when any of the following is in play:

Both tubes flexible. The bending compromise $u = \frac{\sum E_i I_i \mathbf{u}_i^*}{\sum E_i I_i}$ becomes nontrivial — the resulting curvature is some intermediate of the two precurvatures, and changes as the tubes are rotated relative to one another. This is the central CTR steering mechanism in clinical designs (e.g. Dupont, Webster).

Tip wrench. A force $\mathbf{F}_{\text{tip}}$ or moment $\mathbf{M}_{\text{tip}}$ at the distal end propagates back through the moment-balance equation $\mathbf{m}' + \mathbf{p}' \times \mathbf{n} = 0$, deflecting the rod away from its natural shape. The protruding portion is no longer a circular arc; the BVP must be solved numerically.

Distributed load. Gravity on a long thin protruding tube can deflect the tip noticeably, especially in larger-scale CTRs.

Torsional lag. With a sufficiently long flexible tube, the rotation ψ applied at the base does NOT instantly transmit to the tip — the tube "winds up" elastically. The torsional ODE $\theta_i''(s) = (E_i I_i / G_i J_i)(u_x u_{i,y}^* - u_y u_{i,x}^*)$ with appropriate BCs governs this. Significant in clinical CTRs and a known source of control instability ("snapping").

Friction between tubes. Modeled as distributed contact forces; introduces hysteresis and history-dependence in the configuration.

Dynamics. For trajectory planning or feedback control the static BVP is replaced by a PDE in (s, t).

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7. Numerical solution: shooting

The general CTR static problem is a two-point BVP:

- proximal end ($s = 0$): pose $\mathbf{p}(0), \mathbf{R}(0)$ given, base rotations $\theta_i(0) = \psi_i$ given;
- distal end ($s = L$): tip wrench $(\mathbf{n}(L), \mathbf{m}(L)) = (\mathbf{F}_{\text{tip}}, \mathbf{M}_{\text{tip}})$ given.

The unknown initial values are $\mathbf{u}(0)$ and $\theta_i'(0)$. Standard procedure:

1. Guess $(\mathbf{u}(0), \{\theta_i'(0)\})$.
2. Integrate the kinematic + constitutive + equilibrium ODEs from $s = 0$ to $s = L$ (e.g. RK4).
3. Compute residual $\mathbf{r} = (\mathbf{n}(L), \mathbf{m}(L)) - (\mathbf{F}_{\text{tip}}, \mathbf{M}_{\text{tip}})$.
4. Update guess with Newton's method: $\Delta\mathbf{x} = -J^{-1}\mathbf{r}$, where J is the Jacobian of the residual w.r.t. the unknown initial conditions (computed by forward differences or by sensitivity ODEs).
5. Iterate until $\|\mathbf{r}\| < \text{tol}$.

For multi-segment CTRs (different tubes overlapping in different stretches of s), integrate one segment at a time and apply continuity at segment boundaries — moments and positions match,

but curvature can jump where a stiffer tube ends.

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8. Summary table

Concept	Equation
Position kinematics	$\mathbf{p}'(s) = \mathbf{R}(s)\mathbf{v}(s)$
Frame kinematics	$\mathbf{R}'(s) = \mathbf{R}(s)[\mathbf{u}(s)]_\times$
Constitutive (bend+torsion)	$\mathbf{M}(s) = \mathbf{K}_{bt}(\mathbf{u}(s) - \mathbf{u}^*(s))$
Force balance	$\mathbf{n}'(s) + \mathbf{f}(s) = 0$
Moment balance	$\mathbf{m}'(s) + \mathbf{p}'(s) \times \mathbf{n}(s) + \mathbf{l}(s) = 0$
Hollow circular bending stiffness	$EI = \frac{\pi E}{4}(r_o^4 - r_i^4)$
Tube-local vs. common curvature	$\mathbf{u}^{(i)} = R_z(-\theta_i)\mathbf{u} + \theta_i'\hat{z}$
CTR bending compromise	$u_{x,y}(s) = \dfrac{\sum_i E_i I_i \, u^*_{i,(x,y)}(\theta_i)}{\sum_i E_i I_i}$
Torsional ODE per tube	$\theta_i''(s) = \dfrac{E_i I_i}{G_i J_i}(u_x u^*_{i,y} - u_y u^*_{i,x})$
Your case (rigid + flexible, no load)	Inside: $\mathbf{u} = 0$; outside: $\mathbf{u} = \mathbf{u}_2^* \Rightarrow$ circular arc
Tip pose (your case)	$\mathbf{p}_{\text{tip}} = (0, 0, L_o)^\top + R_z(\psi)\,(R(1 - \cos\frac{d}{R}),\ 0,\ R\sin\frac{d}{R})^\top$